

Unit – 4 : Session – 3 : SLO – 2 | Lambda Expressions and Anonymous Functions**I. Programming Task — Custom Higher-Order Function**

1. `def transform(items, func):`
2. `"""Apply func to every element of items and return the new list."""`
3. `return [func(x) for x in items]`
5. `# usage`
6. `nums = [1, 2, 3, 4]`
7. `print(transform(nums, lambda x: x * x)) # [1, 4, 9, 16]`
8. `print(transform(nums, lambda x: x + 10)) # [11, 12, 13, 14]`

II. Explanation

1. `transform()` is higher-order because it accepts another function (`func`) as a parameter.
2. The lambda expressions are anonymous functions defined inline — no `def` or name needed.
3. This pattern generalizes `map()`: any transformation can be plugged in without changing `transform()` itself.